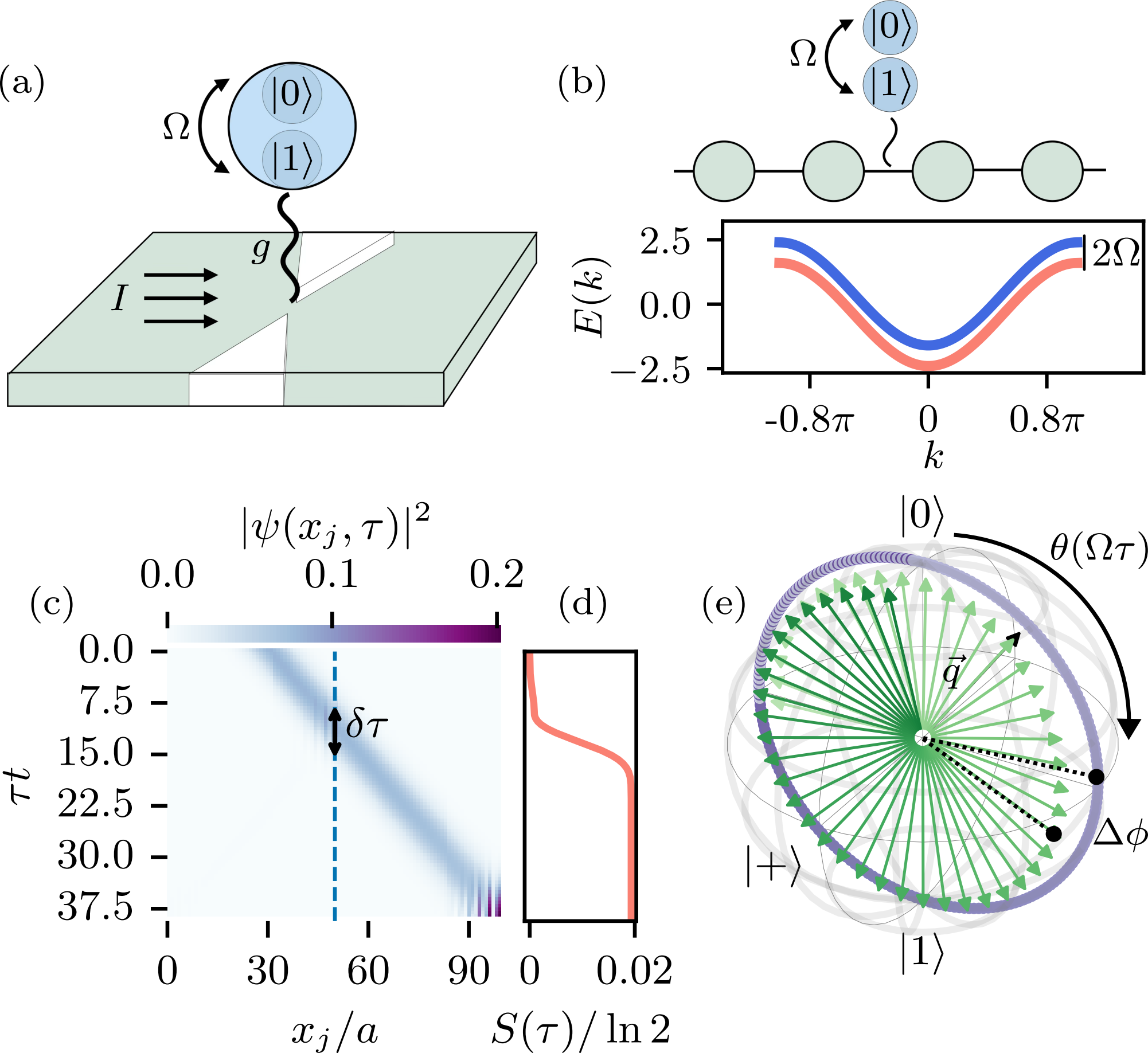

(a)
$|0\rangle$
$|1\rangle$
Ω
g
I
(b)
$|0\rangle$
$|1\rangle$
Ω
$E(k)$
2.5
0.0
−2.5
2Ω
-0.8π
0
0.8π
k
(c)
$|\psi(x_j,\tau)|^2$
0.0
0.1
0.2
τt
0.0
7.5
15.0
22.5
30.0
37.5
$\delta\tau$
0
30
60
90
x_j/a
(d)
0
0.02
$S(\tau)/\ln 2$
(e)
$|0\rangle$
$\theta(\Omega\tau)$
$\vec{q}$
$\Delta\phi$
$|+\rangle$
$|1\rangle$